

Instructions for Authors (Pages 1-4):

Like other scientific journals, the *Journal of Emerging Investigators* (JEI) requires authors to follow specific guidelines when submitting their manuscripts. This makes it easier to process submissions and ensures consistency across all published articles.

We have created a template to guide your writing and help you with formatting your manuscript. You **must use the template** provided on pages 5-14, which includes instructions, examples, and advice for each section. We've also pointed out common mistakes that can delay your submission or will require revision later on.

Our online [submission guide](#) contains more information than presented here. All authors are required to carefully read the online guide and the template **before starting to write** and **again before submitting** to ensure that the manuscript meets JEI's requirements. *This will save you time during the submission process!*

Using The Template:

Your manuscript **must** be organized in the following order:

1. Summary
2. Introduction
3. Results
4. Discussion
5. Materials & Methods
6. Acknowledgments (if applicable)
7. References
8. Figures and Figure Titles/Captions
9. Tables with Titles/Captions
10. Appendix (if applicable)

Do **not** change the order of the sections, add any sections, or remove any sections except for the Acknowledgments and Appendix if not needed.

Do not simply copy and paste your text onto this template. As you write, refer to the information we've provided for each section.

- Any [blue underlined text](#) in the template is linked to the corresponding submission guideline webpage, where additional details can be found.

Formatting Your Manuscript:

General

- **Font:** Arial, size 11, automatic black
- **Line Spacing:** 1.5x
- **Margins:** 1-inch on all sides
- **Headings:** Arial, size 11, automatic black, and **bolded**
 - Do not use Word's automated headings and subheadings.
- Leave the continuous line numbering (numbers to the left of the text) in the document. They are necessary for the scientific review process.
- **Once you finish writing:**
 - Remove any non-heading text (*in italics*) from the template that we provided (i.e., instructions, examples, common mistakes).

- Remove all spaces between sections. For example, the Results heading should appear on the line immediately after the last sentence of the Introduction, etc.
 - The Title Page and Summary should have their own separate pages.
- Delete the “Instructions for Authors” pages (1-3) from the manuscript file.

Pages and Length

- 10-page limit, counted from the start of the **Introduction** to the end of the **Materials & Methods**.
 - At initial submission, you may be 1 to 1.5 pages over the limit. We’ll work with you to refine the material presented during the revision process.

Figures and Tables

- Up to 8 Figures and Tables maximum
- If your manuscript includes symbols, use the “Insert→Symbol” function in Word.
 - Do not copy and paste symbols from other sources, as they will not be recognizable by the system we use (Editorial Manager).

AI Writing Policy

- The use of artificial intelligence (AI) systems to write complete manuscripts or parts of a manuscript is considered [academic dishonesty](#). Any manuscript found to be written with the help of AI will be **returned to authors** and the severity of AI use will determine if we would allow resubmission of the work after editing by the authors.
- Students may use AI to help them understand concepts and techniques in the literature they are citing, but they must take care to **not** use the phrasing provided by the AI tool used in their manuscript.

Plagiarism

- At JEI, it is the responsibility of student authors and their mentors to ensure that they are not plagiarizing.
- All information from external sources should be properly paraphrased. Note that proper paraphrasing is more than just switching or replacing a few words or rearranging them.
- JEI also does not permit the use of direct quotations from other sources, even if you have cited the source.
- Definitions from external sources will also need to be properly paraphrased.
- Scientific terms, names, organizations, and acronyms do not need to be paraphrased.

Common Mistakes

- **Bulleted lists:** Bulleted, numbered, or lettered lists should **not** appear in the main text (i.e., Summary through Materials and Methods).
- **Singular first person tense:** Because JEI requires more than one author to be listed on all manuscripts, all sections should use the plural first person tense (e.g., We performed the experiment) and **not** singular first-person tense (e.g., I performed the experiment)
- **Present tense:** Because you have already completed your experiments, your manuscript should be in past tense (e.g., We observed XYZ) and **not** in present tense (e.g., We observe XYZ).
- **Passive voice:** Active voice (e.g., We did the experiment at the school) should be used throughout the manuscript, **except** the Materials & Methods sections where passive voice should be used (e.g. The experiment was done at the school.)
- **Formatting as a finished article:** Template formatting instructions should be followed in this document and should not be formatted in the same way as articles on our website. In

other words, there should be **no** columns, **no** article-specific text coloring, **no** figures/tables in the main text, etc.

Final Checks Before Submitting:

- ☐ All authors have given permission for this work to be published at *JEI*.
- ☐ All images and tables must be original (your own work):
 - **Any image and/or table published in other sources requires approval for copyright reasons.** Reach out to us before submission if you are considering using a previously published image or table.
 - If you used software like Biorender that requires a license to publish generated figures, **a copy of the license must be uploaded to the File Inventory in Editorial Manager.** Note that the free version of Biorender does not allow for the publication of generated figures.
- ☐ All study approval documents (if applicable) are ready to be uploaded .
 - See our [Human and Animal Subjects guidelines](#) for more information.
 - If your study involves vertebrate animals or human participants, approval should be obtained **prior** to starting your study.
 - For consent forms, only upload the blank version that you gave to participants.
 - If we need to see filled-out forms, we will contact you.

Using Editorial Manager

- The adult senior author or parent/guardian must create an account:
 - The senior author oversees the research. The corresponding author is whoever submits the manuscript, whether it be the senior author or the parent/guardian.
 - If a parent/guardian is submitting the manuscript and is not the senior author, ensure the senior author has given permission to publish the manuscript. The senior author will also need a valid email address in Editorial Manager, in the “Other Authors” section, if they are not serving as the corresponding author.
 - The adult who submits the work will be the one who receives emails from JEI regarding the submission. This does **not** reflect the actual order of authors listed on the manuscript. The order of authors on the final publication will be taken from the Word document.
 - Student authors can include their email address in the “Other Authors” section to be copied on all journal correspondence.
 - **Under no circumstances** is the student author allowed to submit their manuscript to *JEI* using their own Editorial Manager account. Students are also not permitted to have access to their parent’s/adult corresponding author’s email accounts or Editorial Manager accounts.
- The manuscript file must be uploaded as a Word document.
- **Figures must be saved as JPG, PNG, or TIFF files to be recognized by the system. PDF files for figures will not be accepted.**
 - **Figures should be uploaded separately in Editorial Manager as well as placed at the end of the template above their titles/captions.**
 - **If a figure has multiple panels (e.g., Figure 1a, b, c), it must be uploaded as a single file (e.g., Figure 1).**

- Tables should **not** be uploaded as image files but rather included at the end of the manuscript as Word-generated tables using the “Insert→Table” function.
- Any study approval documents should be uploaded under “Other”.
- Review our online [submission checklists](#) for all other items needed to [submit](#).
 - We also have a [guide](#) and [video](#) that explain how to use Editorial Manager.

Contacting JEI:

Should you have any questions throughout the submission process, please contact us at submissions@emerginginvestigators.org.

- Both students and adult authors are permitted to contact JEI with their questions. If students have a question for JEI staff, they should cc their adult corresponding author/senior author on the email.
- If you have submitted already, include the title, and, if accepted for scientific review, the manuscript number (e.g., JEI-22-153) in any email to the journal.
- If replying back to a JEI staff member’s email, ensure you “**reply all**” when responding to prevent communication delays, as our submissions inbox is monitored by multiple volunteers.

Article Title (110 characters or less)

First Author¹, Second Author², Third Author³

¹ Department (if applicable), Institution/School, City, State, Country (if non-US)

² Department (if applicable), Institution/School, City, State, Country (if non-US)

³ Department (if applicable), Institution/School, City, State, Country (if non-US)

Student Authors

First Student Author, Middle/High School

Second Student Author, Middle/High School

KEYWORDS: Accessibility, Wearables, Stereoscopy, Audio, Transformers

OVERVIEW: Provide a 2- to 3-sentence overview of your manuscript. This will be used as a blurb for your article when posting the final publication on our website.

SUMMARY

A short summary of the whole manuscript, including:

- ☐ The problem that prompted the study or why the study is valuable
- ☐ The research question or purpose of the study
- ☐ The main hypothesis (specifically, "We hypothesized...")
- ☐ Summary of the results with a very brief overview of the methods for context
- ☐ Key takeaways or conclusions and implications of the findings

Common mistakes seen:

- Inclusion of in-text reference citations. This is the only section of the manuscript where in-text reference citations should not be included.
- Formatting as multiple paragraphs. This section should be in a narrative format, ideally as a single paragraph. Bulleted, numbered, or lettered lists should also **not** appear in the main text (i.e., Summary through Materials and Methods).

INTRODUCTION

We recommend that Introductions follow an inverted pyramid format, where you start with broad facts and get more specific with each paragraph about your research question.

Your Introduction should:

- ☐ Describe the overarching scientific topic of the study – Describe primary, all-encompassing concept or theme that ties together several smaller parts or ideas of the project or study

- Provide background information on the research question so readers understand why this question is being asked AND is of interest
- Have numbered, parenthetical in-text [reference citations](#) throughout
 - This section is often the one with the most citations.
- End with a brief concluding paragraph that states:
 - A clearly stated hypothesis of the study (specifically, “We hypothesized...”)
 - Major results
 - 1–2 key takeaways (conclusions)

Common mistakes seen:

- Presenting work that is not hypothesis-driven. See our [guidelines](#) for more information.
 - If your study involves an invention, machine learning model, or a methods optimization, see our [specific guidelines](#) for those types of studies.
- Not having any or enough citations to support claims made. *Any claim made that is not considered “common knowledge” must have a citation provided. For example, if you are stating a fact, such as “1.3 million people die from tuberculosis every year”, include an in-text reference citation to support that claim.*
- Presenting hypotheses as a null and an alternative hypothesis. When writing a manuscript, you don’t want to have text that could potentially confuse readers, so you should just state the question you are interested in testing (which is often the alternative hypothesis).
- Including subsections/subheadings. This section should be in a narrative format with thoughtful transition sentences between paragraphs. **No subheadings should be included. Bulleted, numbered, or lettered lists should not appear in the main text (i.e., Summary through Materials and Methods).**

RESULTS

The results are where you report the observations from your experiment(s). For each experiment:

- Describe the rationale (why does this experiment make sense to do?)
- Briefly explain how the experiment was done
 - This is to help your readers understand the data. Full details will be provided in “Materials and Methods.”
- State what you observed in your experiment(s), *making sure you reference any figures or tables that contain your results*
 - *Your findings should be as detailed as possible and state exact measurements. For example, if your experiment involved measuring plant growth in inches, state*

the observed growth in average inches instead of saying you saw an increase/decrease in growth.

- Exact measurements and p-values should be reported in this section.

The Results can be divided into subsections with a header (in italics) summarizing the content of that subsection. This is particularly helpful in keeping information organized if you have different experiments.

Common mistakes we see are that present tense instead of past tense is used to describe results and too much focus is placed on figures, statistics, and/or numbers without putting them in the larger context of the experiment. We recommend that authors think about referencing these items as they do when citing literature to support claims. An example is given below:

Incorrect:

In Figure 1, we can see that with a p-value of 0.01 seeds planted in soil containing 40 ppm nitrogen grew significantly taller than those in soil with 20 ppm nitrogen.

Correct:

*Seeds planted in soil containing 40 ppm nitrogen grew significantly taller than those in soil with 20 ppm nitrogen (1.7 ± 0.3 in vs. 1.0 ± 0.1 in; $p = 0.01$, two-sample t-test, **Figure 1**).*

DISCUSSION

The discussion is often the longest section of a manuscript. This section is where you want to discuss your results and interpret what these results mean.

Your discussion should:

- ☐ Summarize the experimental results and draw conclusions from these results
 - Results must be mentioned in the results section first.
- ☐ Discuss factors or limitations that could have influenced your results
 - Human error is assumed, so it should not be the primary focus of this section.
 - Think about whether your experiments should have had more replicates, different controls, or if there were better ways to measure your data than what you did.
- ☐ Tell the readers about the significance/impact of your results
 - Be careful not to overinterpret your results!

- *Example of overinterpretation:*
 - *Treating cells in vitro and seeing an effect (i.e., decrease in glucose) does not indicate that you have found a new compound that will work in animals or humans in vivo to treat a disease (i.e., diabetes).*
- *Also, remember that correlation does not equal causation.*
- *Your data may support a hypothesis, but they cannot definitively prove it. Avoid words like “prove” in your interpretation.*
- *Discuss remaining scientific questions and potential future studies*
- *Conclude with a paragraph that briefly summarizes your results and touches on the overall impacts of your study*

Common mistakes seen:

- *Short discussion sections. This section should be a significant portion of your manuscript and not just 1–2 paragraphs.*
- *Including subsections/subheadings. This section should be in a narrative format with thoughtful transition sentences between paragraphs. **No subheadings should be included.***
- *Focus is on limitations of study rather than the larger implications. While limitations should be discussed, they should not take up the majority of the writing in this section.*

MATERIALS AND METHODS

*This is usually the most challenging part of the manuscript for our student authors to write, as it differs the most from what you would do in a school lab report or science fair project. You want to provide enough details for someone to replicate your experiments, but you should **not** list everything in a step-by-step fashion.*

*The Materials and Methods can be divided into subsections with a header (in italics), summarizing the content of that subsection. This is particularly helpful in keeping it organized if you have different experiments. **However, materials should not be a separate section but should appear in the context of the methods used for your study. A company and catalog number should be specified for any unique materials, equipment, and animal strains. There is no need to mention where generic items were obtained (i.e., a test tube).***

If you had any code that was self-generated for your manuscript, then you can either upload it to a GitHub page (if you have one) and provide an in-text reference citation for your link or include the full script or Github link as an appendix to your manuscript.

If equations are central to your methods, ensure they follow our journal's [formatting guidelines](#).

We have provided an example below of a step-by-step protocol and how it was changed into a protocol for a manuscript:

Incorrect:

- 1) Grow bacteria up overnight with shaking at 30°C in 2 mL of broth in a glass test tube*
- 2) Perform a serial dilution on the overnight cultures (1:10 to 1:1000) in sterile water*
- 3) Using a sterile cell spreader, spread out 50 uL of each dilution onto blood agar plates.*
- 4) Incubate at 30°C overnight.*
- 5) Count the resulting colonies and categorize based on hemolysis of the blood agar.*

Correct:

Bacteria samples in glass tubes containing 2 mL LB broth (Sigma, Cat# L2542) were grown at 30°C overnight with shaking in the SD-ura incubator. Overnight cultures were diluted via serial dilution from 1:10 to 1:1000 in sterile deionized water. Using a cell spreader, 50 uL of each dilution spread onto tryptic soy agar (TSA) plates (GrowingLabs, Cat# G60) to ensure even coating. Then, plates were inverted and incubated at 30°C overnight. The resulting colonies were counted.

ACKNOWLEDGMENTS (Optional)

*Authorship is considered the highest form of acknowledgement in scientific writing. **This means that anyone in your author list can't appear in your acknowledgements.** If anyone or anyplace helped you with your research but didn't meet the requirements for authorship, mention them in this section. If you received any funding for your work, this is where you want to acknowledge it!*

REFERENCES

- Every effort should be made to use primary sources or reputable secondary sources. Try to avoid/limit referencing Wikipedia or blogs and social media posts.*

- References are numbered based on the order in which they are cited in the text. For example, the first reference you cite should be numbered (1) at the end of the sentence. The reference corresponding to this in-text reference citation will be listed as the first reference in a numbered list in the References section. If using a citation manager, style files are available in our [Author Resources](#).
- *Hyperlinks (the link to the webpage) should be included with "https://" for any reference with a weblink or DOI.*
- *References with three or more authors should list only the first author, followed by "et al."*
- *No hanging indentation should be used.*
- See [previously published articles](#) to see how Reference lists should be formatted.

Example formats:

Scientific journal articles (only italicize the journal name)

Author name(s). "Title of Article." Name of Journal, vol. X, no. X, Day Month Year, pp. XX-XX.
<https://doi.org/>

Websites (only italicize the website name)

Author name(s) (if any). "Title of Page or Article." Website Name. <https://webpage url>. Accessed Day Month Year.

Book (only italicize the book title)

Author names(s). "Chapter Name." Title of Book, edited by Editor name(s), Xth ed., vol. X, Publisher, Year, pp. XX-XX.

Figures and Figure Titles/Captions

Place your figures in this section with their bolded title and un-bolded caption located below the figure.

Figure captions are often one of the hardest pieces of a manuscript to write. The basic elements of a caption are as follows: **title**, **what is shown**, **methods**, **statistical tests and values** (if applicable), and **replicates**.

Example caption:

Figure 1. Herbal formulation (HF1) eliminated PMA-induced overexpression of COX-2 in MCF-7 cells. Bar graph showing mean $\pm$ SD COX-2 mRNA fold change ($n=3$). MCF-7 breast

cancer cells were grown under either control conditions, in 10ng/mL PMA (COX-2 overexpression), or in 10 ng/mL PMA + 0.3 mg/mL HF1 for 24hrs. One-way ANOVA, *** $p < 0.001$.

Note: Highlighting provided is for demonstration purposes only and should **not** be included in your figure/table titles and captions. Additionally, your figure or table captions should capitalize the first letter of the first word and then only proper nouns and acronyms after that.

Tables with Titles/Captions

Tables should be placed **above** their bolded title and un-bolded caption. Make sure your captions explain any abbreviations used in the table. Any references provided in the table should also be cited somewhere in the manuscript and given in the table as: Last name (year).

Tables should be in an editable format (i.e., using the “Insert→Table” function in Word (**not** provided as JPEG, PNG, or TIFF files) and efforts should be made to limit them to one page in size. If tables cannot be made to fit on one page, while still being legible, they will be placed in the appendix for the manuscript.

Note: Only a maximum of 8 figures and tables total is allowed for JEI articles. Supplemental figures/tables should not be added unless explicitly necessary for understanding the manuscript. If adding supplementary figures/tables, include them in the Appendix in addition to a statement justifying their inclusion.

Appendix (If applicable)

JEI allows appendices on a case-by-case basis depending on the type of research being presented.

If you feel your work requires an appendix, please include it along with a separate statement of why this information is necessary to include as an appendix rather than in the actual manuscript.

Note: Code/programming script or Github links do not need any justification statement provided.